

**SAPTHAGIRI COLLEGE OF ENGINEERING, BENGALURU-560 057.**  
**(Affiliated to VTU, Belagavi & Approved by AICTE, New Delhi)**  
**DEPARTMENT OF MATHEMATICS**  
**QUESTION BANK (III SEM)**  
**ACADEMIC YEAR: 2024-25 (ODD SEMESTER)**



**MODULE-2:**

**CURVE FITTING, CORRELATION AND REGRESSIONS**

**Sub: MATHEMATICS-III**

**Sub Code: BMATE301**

1. Fit the least square straight line  $y = ax + b$  to the following data

|   |     |     |      |      |      |     |     |
|---|-----|-----|------|------|------|-----|-----|
| x | -5  | -3  | -1   | 0    | 1    | 2   | 4   |
| y | 0.4 | 0.1 | -0.2 | -0.3 | -0.3 | 0.1 | 0.4 |

2. Find a least square straight line for the following data:

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| x | 1 | 2 | 3 | 4 | 5 | 6 |
| y | 6 | 4 | 3 | 5 | 4 | 2 |

3. Fit a straight line of the form  $y = a + bx$  to the following data.

|   |    |    |    |    |    |
|---|----|----|----|----|----|
| x | 1  | 2  | 3  | 4  | 5  |
| y | 14 | 27 | 40 | 55 | 68 |

4. Fit a second degree parabola of the form  $y = a + bx + cx^2$  to the following data

|   |   |     |     |     |     |
|---|---|-----|-----|-----|-----|
| x | 0 | 1   | 2   | 3   | 4   |
| y | 1 | 1.8 | 1.3 | 2.5 | 2.3 |

5. Fit a curve of the form,  $y = ax^2 + bx + c$  to the following data:

|   |    |    |   |    |    |
|---|----|----|---|----|----|
| x | 1  | 2  | 3 | 4  | 5  |
| y | 10 | 12 | 8 | 10 | 14 |

6. The following table gives the heights of Father(x) and Sons(y):

|   |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|
| x | 65 | 66 | 67 | 67 | 68 | 69 | 70 | 72 |
| y | 67 | 68 | 65 | 68 | 72 | 72 | 69 | 71 |

Find the lines of regression and hence calculate the co-efficient of correlation.

7. Find the curve at best fit of the form  $y = ax^6$  to the following data.

|   |     |   |     |   |      |
|---|-----|---|-----|---|------|
| x | 1   | 2 | 3   | 4 | 5    |
| y | 0.5 | 2 | 4.5 | 8 | 12.5 |

8. Calculate the coefficient of correlation and obtain the lines of regression for the following data:

|   |   |   |    |    |    |    |    |    |    |
|---|---|---|----|----|----|----|----|----|----|
| x | 1 | 2 | 3  | 4  | 5  | 6  | 7  | 8  | 9  |
| y | 9 | 8 | 10 | 12 | 11 | 13 | 14 | 16 | 15 |

9. From the following data of the ages of husbands and wives

|   |    |    |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|----|----|
| x | 36 | 23 | 27 | 28 | 28 | 29 | 30 | 31 | 33 | 35 |
| y | 29 | 18 | 20 | 22 | 27 | 21 | 29 | 27 | 29 | 28 |

Find the two regression lines and calculate the husband age when the wife is 16- year-old

10. Ten participants in a dancing contest were ranked by two judges as given in the following table

|   |   |   |   |    |   |   |   |    |   |   |
|---|---|---|---|----|---|---|---|----|---|---|
| x | 1 | 6 | 5 | 10 | 3 | 2 | 4 | 9  | 7 | 8 |
| y | 6 | 4 | 9 | 8  | 1 | 2 | 3 | 10 | 5 | 7 |

Find the rank correlation coefficient.

11. The revolution (r) and time (t) are related by quadratic polynomial  $r = at^2 + bt + c$ . Estimate number revolution (r) for time(t) 3.5 units. Given

|            |     |     |     |     |     |     |    |
|------------|-----|-----|-----|-----|-----|-----|----|
| Revolution | 5   | 10  | 15  | 20  | 25  | 30  | 35 |
| Time       | 1.2 | 1.6 | 1.9 | 2.1 | 2.4 | 2.6 | 3  |

12. Ten competitors in music contest are ranked by three judges A, B & C in the following order.

|   |   |   |   |    |   |    |   |    |   |   |
|---|---|---|---|----|---|----|---|----|---|---|
| A | 1 | 6 | 5 | 10 | 3 | 2  | 4 | 9  | 7 | 8 |
| B | 3 | 5 | 8 | 4  | 7 | 10 | 2 | 1  | 6 | 9 |
| C | 6 | 4 | 9 | 8  | 1 | 2  | 3 | 10 | 5 | 7 |

Use rank correlation coefficient to decide which pair of judges have the nearest approach to common taste of music.

13. In a partially destroyed lab record, only the lines of regression of  $y$  on  $x$  and  $x$  on  $y$  are available as  $4x - 5y + 33 = 0$  and  $20x - 9y = 107$  respectively. Find (i) Mean value of  $x$  and  $y$   
(ii) SD of  $y$   
(iii) co-efficient of correlation between  $x$  and  $y$ .

14. Find the regression lines and hence find the coefficient of correlation for the following data.

|   |    |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|----|
| x | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |
| y | 12 | 11 | 13 | 15 | 14 | 17 | 16 | 19 | 18 |

15. Determine the rank correlation for the following data which shows the marks obtained in two quizzes in mathematics.

|                       |   |   |   |    |   |   |    |   |   |   |
|-----------------------|---|---|---|----|---|---|----|---|---|---|
| Marks in first quiz X | 6 | 5 | 8 | 8  | 7 | 6 | 10 | 4 | 9 | 7 |
| Marks in first quiz Y | 8 | 7 | 7 | 10 | 5 | 8 | 10 | 6 | 8 | 6 |

16. Find the rank correlation for the following data.

|   |     |     |     |     |     |     |     |     |     |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| x | 56  | 42  | 72  | 36  | 63  | 47  | 55  | 49  | 38  | 42  | 68  | 60  |
| y | 147 | 125 | 160 | 118 | 149 | 128 | 150 | 145 | 115 | 140 | 152 | 155 |

17. Ten students got the following ranks in two subjects:

|         |   |   |    |   |   |    |   |   |   |   |
|---------|---|---|----|---|---|----|---|---|---|---|
| Maths   | 3 | 8 | 9  | 2 | 7 | 10 | 4 | 6 | 1 | 5 |
| Physics | 5 | 9 | 10 | 1 | 8 | 7  | 3 | 4 | 2 | 6 |

Calculate the rank correlation coefficient.

18. The regression lines between the height ( $x$ ) in inches and weight ( $y$ ) in lbs. of male students are  $4y - 15x + 530 = 0$ ,  $20x - 3y - 975 = 0$

Find i) Mean height and weight of students

ii) Coefficient of correlation between height and weight.

iii) Estimate the weight when the student has a height of 80 inches.

19. If  $\theta$  is the angle between the two regression lines, show that  $\theta = \left( \frac{1-r^2}{r} \right) \frac{\sigma_x \sigma_y}{(\sigma_x^2 + \sigma_y^2)}$ .

Indicate the significance of the cases  $r = 0$  and  $r = \pm 1$

20. Find the correlation co-efficient between  $x$  and  $y$  from the following data

|   |     |     |     |     |     |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|-----|
| X | 78  | 89  | 97  | 69  | 59  | 79  | 68  | 57  |
| Y | 125 | 137 | 156 | 112 | 107 | 138 | 123 | 108 |

21. Find the correlation coefficient between  $x$  and  $y$  when the lines of regression are  $2x - 9y + 6 = 0$  and  $-2y + 1 = 0$ .

22. Find the regression line of  $y$  on  $x$  for the following data

|   |   |   |   |   |   |   |    |    |
|---|---|---|---|---|---|---|----|----|
| X | 1 | 3 | 4 | 6 | 8 | 9 | 11 | 14 |
| Y | 1 | 2 | 4 | 4 | 5 | 7 | 8  | 9  |

Estimate the value of  $y$  when  $x=1$

